

Trending Open Source Technologies

Open source technologies are in a constant state of flux and evolution. This article is meant to draw attention to emerging and trending open source technologies, giving their salient features in a nutshell.

#trending

Open source technologies are evolving fast with the continuous development of innovative solutions. This article mainly focuses on describing a few such breakthrough technologies and tools under the open source umbrella.

Open source in virtual reality and augmented reality

Virtual reality is one of the key technologies that goes way beyond gaming into fields like nuclear reaction simulation, flight/battlefield simulation, and phobia treatment through the use of virtual models, etc. A few noteworthy open source tools are OSVR, ARToolkit, OpenSimulator, etc.

- **OSVR or Open Source Virtual Reality** (<http://www.osvr.org/>) is an open source project initiated and sponsored by Sensics and Razer. It is provided under the Apache 2 licence. The main components include open source hardware (HMD or head mounted display) and an OSVR framework, which is open source software that provides a standardised way of discovering and configuring various VR and AR peripherals.
- **ARToolkit** (<http://artoolkit.org/>) is an open source C library that aims at developing augmented reality applications. The open source project has its source code hosted over GitHub. It's a cross-platform tool with support for Windows, Mac, Linux, iOS and Android platforms and hence is easily portable. The software makes use of various types of computer vision for robust feature tracking.
- **OpenSim** (<http://opensimulator.org/>) is an open source multi-platform application server (mainly written in C#), which is primarily used for simulating virtual environments. The source code is released under a BSD licence, such that it can be easily used in projects. Still under development, this .NET based framework allows many features and an inheritance based approach that makes it highly extensible.

Open source in machine learning

Among the most used open source tools/libraries for building efficient ML applications are Apache Mahout, Weka, Scikit-Learn, etc. Two of the most recent ML engines that have been made open source by the developer organisations are

TensorFlow from Google and DMTK (Distributed Machine Learning Toolkit) from Microsoft.

- **Tensor Flow** (<https://www.tensorflow.org/>) is a numerical library mostly used for deep neural network based computation, where all the computations are modelled as data flow graphs. Recently outsourced by Google, the main advantages of this tool are its Python based API and easy-to-use C++ interface, and its flexibility to run on a variety of hardware, ranging from cheap commodity hardware to heterogeneous GPU based systems.
- **DMTK** (<http://www.dmtk.io/>) is an open source machine learning toolkit developed by Microsoft researchers. The toolkit provides a framework that can be used to train machine learning models on Big Data. Following a client-server based model, this toolkit also provides two inter-process communication libraries, MPI and ZMQ, to handle a variety of cluster environments used for model training.
- **Scikit-learn** (<http://scikit-learn.org/>) is a Python based tool with very active community support. The underlying compiled libraries NumPy, SciPy, Matplotlib, etc, make the tool fast and efficient. It is used for both academic as well as commercial ML development due to its versatile and simple APIs.

Open source in IoT

The IoT (Internet of Things) is about how smart devices can collect and exchange data that is transmitted over the Internet via other connected devices and sensors. IoT is the most talked technology around the world. Due to its advantages and popularity, most companies are moving towards solutions based on it. Considering how businesses can take advantage of open source IoT, one can check out the following options.

Development tools:

- **Arduino** is one of the most preferred tools for IoT since it provides both a hardware specification and software with an IDE programming language for IoT. Refer to <https://www.arduino.cc/> for more information on getting started with Arduino.
- **Node-RED** built on *node.js* (<http://nodered.org/>) provides visual tools for developers to connect together hardware devices, APIs and services for the Internet of Things.